Valuation Scheme QP CODE: 25900902

MAHATMA GANDHI UNIVERSITY, KOTTAYAM MGU-UGP (HONOURS) Regular EXAMINATION November 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCMOS100 - PC HARDWARE AND SMARTPHONE TROUBLESHOOTING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 mark

1. c) Install expansion cards [CO1]	6. b) Longer transfer time [CO1]
2. a) Unstable power supply or faulty capacitors [CO2]	7. b) To permanently store the operating system and user data [CO3]
3. d) To accurately detect weak satellite signals for precise location services [CO3]	8. b) Failure to boot or system crashes [CO2]
4. b) Provide continuous power during outages [CO1]	9. d) To perform hardware initialization and provide runtime services [CO2]
5. b) To minimize the risk of solving the wrong problem [CO2]	10. b) RF Section [CO3]

Part B
Short Essay Questions (4*5=20)

11. **Explain the role of the Audio IC in a smartphone and how it ensures high-quality sound.**

Converts digital audio signals to analog & vice versa, Ensures noise-free processing, Controls microphone, speaker, headphone output, Manages sound amplification, Maintains high-quality audio performance **[CO3]**

12. **What type of connector is typically used to attach audio jacks to the motherboard?**

3.5 mm TRRS jack connectors, Uses soldered surface-mount connectors, Sometimes FPC (Flexible Printed Cable) connectors are used to connect audio modules, Provides stable analog sound connection **[CO1]**

13. **Explain the Power-On Self-Test (POST) process and its significance in the boot sequence of a computer.**

POST = Power-On Self-Test, Runs when the computer starts, Checks CPU, RAM, motherboard, keyboard, BIOS, display, Detects hardware failures early, Gives beep codes/messages for troubleshooting **[CO2]**

14. **What is the primary function of storage in a smartphone, and what types of memory are commonly used?**
Stores OS, apps, user data, cache, media, Non-volatile storage, Types used: eMMC, UFS, NAND Flash, RAM functions separately for temporary storage **[CO3]**

15. **Discuss the different types of RAM and their significance in a computer system.**
SRAM: Used in cache memory, fast and expensive, DRAM: Used as system RAM, SDRAM, DDR, DDR2/DDR3/DDR4: Synchronized with system clock, Importance: temporary data storage, system speed, multitasking **[CO1]**

16. **What are the safety precautions to be followed while testing SMPS voltage using a multimeter?**
Disconnect from mains before opening, Check for swollen capacitors or burn marks, Use insulated probes, Do not touch components while powered, Ground yourself to avoid static discharge, Measure DC output carefully **[CO2]**

Part C
Essay Questions (2*10=20)

17. **Explain how Recovery Mode can assist in resolving software issues on a smartphone. Provide examples of tasks that can be performed in this mode.**
What is Recovery Mode, Accessing options like: Factory reset, Cache partition wipe, Apply updates, Reinstall firmware, How it fixes software issues, Examples: boot-loop repair, virus removal, system instability, Importance in troubleshooting **[CO3]**

18. **Explain the function of an SMPS in a computer system. Discuss the significance of the different wire colors in the ATX power supply connector, and how they correspond to specific voltage outputs and connections.**
SMPS function: Converts AC to DC, Provides regulated voltages: +12V, +5V, +3.3V, Protects against over-voltage, over-current, Stable power to motherboard, Wire color meanin**gs:** Yellow: +12V, Red: +5V, Orange: +3.3V, Black: Ground, Purple: +5VSB, Green: PS-ON, Grey: Power Good (Explain purpose of each) **[CO1]**

19.What is a personal computer? What are the different generations of computers? Explain briefly.
Definition: General-purpose digital computer for personal use
Generation**s:**1st Gen – Vacuum tubes, 2nd Gen – Transistors, 3rd Gen – ICs, 4th Gen – Microprocessors, 5th Gen – AI-based systems**.**
Explain characteristics of each **[CO1]**

20. Examine the challenges that users face when interpreting POST indications and the implications for hardware troubleshooting.
Users may not understand beep codes, Different motherboard manufacturers use different codes, Errors may be vague (e.g., memory error but not exact module), Hard to differentiate RAM/CPU/motherboard issues, Missing display makes troubleshooting harder, Importance: accurate diagnosis reduces time & cost. **[CO2]**